# Pranav Tripathi

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## EDUCATION

**University of California, Berkeley | Mechanical Engineering BS | Regents' & Chancellor's Scholar** May 2028
4.0 GPA | Courses: Mechanics, Python & MATLAB, CAD/CAM, Data Structures, Calculus, Linear Algebra

## EXPERIENCE

**Frostbyte Technologies** | *Co-Founder* June 2024 – Present
- Leading my startup team to develop an autonomous robot to clear ice from sidewalks and improve pedestrian safety
- Gained support of Samsung USA ($12,000 award) and Gates Foundation for development and policy compliance
- Designing plow & salt dispenser with 1200 $ft^2$ coverage, FEA for 50+ lbs loads, & TopOpt for 18% weight reduction
- Validating design via Chrono physics engine, and implementing navigation via RTK GPS & LiDAR SLAM on custom PCBs

**D&Y Nexus Labs** | *Mechanical Engineering Consultant* August 2026 –November 2026
- Consulted on the design of an autonomous methane flux chamber for safe and accurate methane emission monitoring
- Reinforced chamber for IP54 water resistance through gaskets/seals, added pressure-equalization to mitigate humidity and temperature variations (up to 30° C), and assisted with an IoT server for monitoring (<5 second latency)
- Converted designs into tolerance drawings and maintained bill of materials (BOM) to stay within budget of $400

**San Francisco Engineers Without Borders (EWB)** | *Engineering Volunteer* August 2026 – Present
- Collaborating with EWB to build fluid systems that provide clean water to 6200+ residents of Tranquillo, Panama
- Expanding current water infrastructure by adding solar-powered pump designs for an extra 7,000 L/hr water supply
- Improving retaining walls and pipeline crossings to sustain loads up to 5 tons, and to have a lifespan of 25 years

**Computer Science Council** | *President* June 2023 – June 2026
- Led 38-member team to organize 14 free computer science competitions and workshops for young local students
- Managed 3 teams (event logistics, advertising, and activity development), to reach over 1,800 students in two years

## RESEARCH

**Embodied Dexterity Group, General Motors (GM)** | *Undergraduate Design Lead* August 2025 – Present
- Leading the QUAD project, a robotic hand with 11 DoF and visuotactile sensing, under PhD Ben Davis and GM
- Designing QDD linkages for high dexterity and an asymmetric underactuated linkage for rigidity and complex grasping, with kinematic and geometric constraints that mimic the size, shape, and movement of human hands
- Improving DFM through 2D metal parts, with 70% faster production time and higher precision (0.2 mm tolerances)

**Boston University Robotics Lab** | *Mechanical Engineering Intern* June 2025 – August 2025
- Created autonomous path-following algorithms that use solely visual input from a camera, under Prof. Roberto Tron
- Developed NumPy simulations that tracked the optical flow of points along a path boundary using inverse kinematics
- Programmed an OpenCV algorithm that calculates deviation from a path by comparing the optical flow of different points, and uses ROS2 feedback nodes and PID for course-correction, resulting in <12% positional error

**Lake Washington Tech Math Dept.** | *Mathematics Research Assistant* December 2025 – May 2026
- Optimized popular search algorithms to enable pathfinding via minimal computations under Prof. Choudhury
- Simulated shortest-path algorithms Dijkstra's and A* in Python using data structures such as queues and linked lists
- Created a novel method that searches from both ends and draws them together, improving efficiency by 52% over A*

## SKILLS

**Mechanical:** 3D Modelling, Kinematics, FEA, Machining, GD&T, DFM/DFA, Microsoft Office (Excel Certified)
**Electronics:** Programming (Python, Java, C++, MATLAB), Github, Circuitry & Sensors (Fritzing)
**Robotics:** ROS2, Computer Vision (Optical Flow, Camera Projections), SLAM, Controls (PID, Odometry), Signals